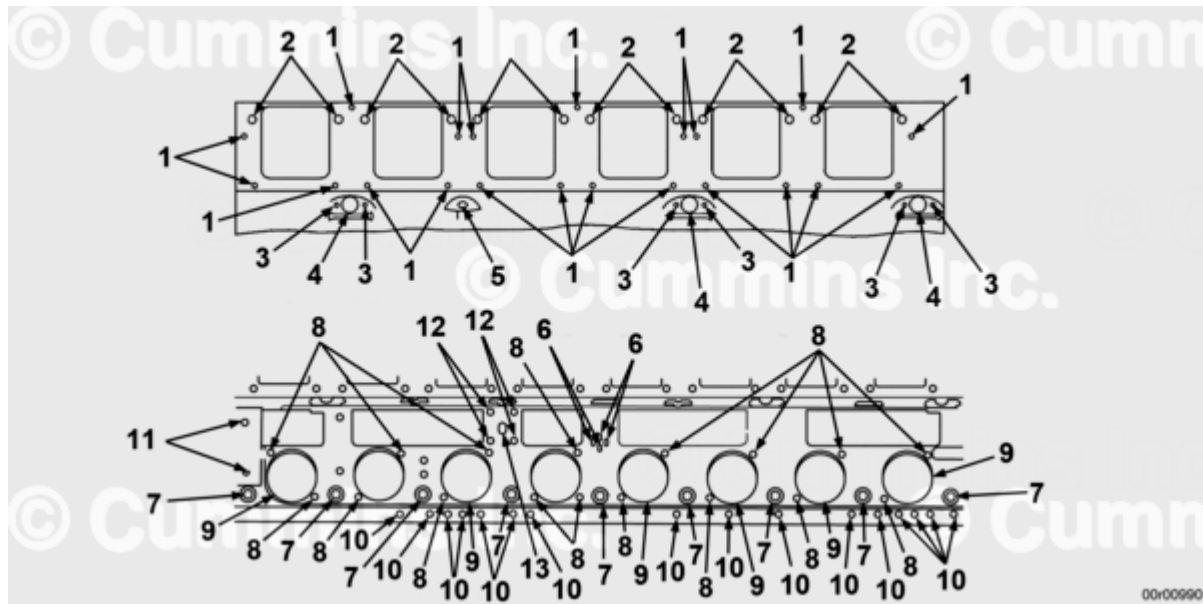
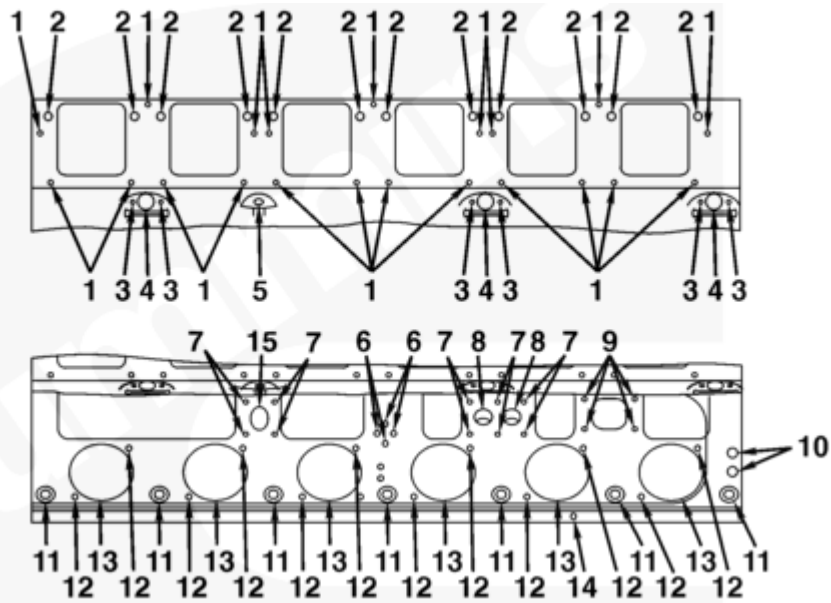


General Information



Left Bank

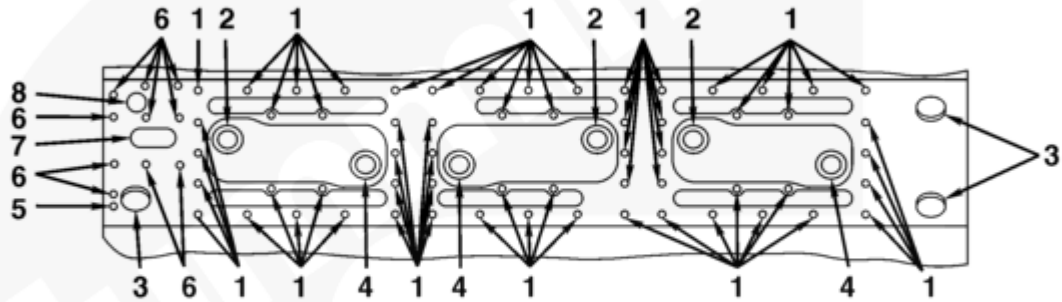
1. Cam cover mounting capscrew holes
2. Cam follower assembly mounting capscrew holes
3. Block heater coolant passage capscrew holes
4. Coolant passage for block heaters
5. Coolant passage to the air compressor
6. Electronic control valve assembly mounting capscrew holes and fuel drillings
7. Main bearing cap side capscrew mounting holes
8. Hand hole cover mounting capscrew holes
9. Hand hole cover holes
10. Fuel filter head mounting capscrew holes
11. Idler shaft oil rifles
12. Fuel pump oil supply blanking plate capscrew holes
13. Fuel pump oil supply passage.



01600281

Right Bank

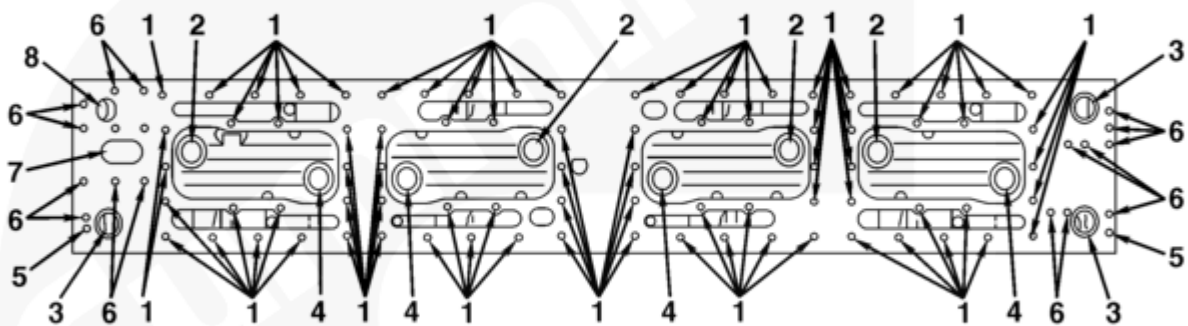
1. Cam cover mounting capscrew holes
2. Cam follower assembly mounting capscrew holes
3. Block heater coolant passage capscrew holes
4. Coolant passage for block heaters
5. Coolant passage to the air compressor
6. Electronic control valve assembly mounting capscrew holes and fuel drillings
7. Oil filter head mounting capscrew holes
8. Oil rifle from oil pump to oil filter head to main oil rifle
9. Water pump outlet connection mounting capscrew holes
10. Idler drillings oil passages
11. Main bearing cap side capscrew mounting holes
12. Hand hole cover mounting capscrew holes
13. Hand hole cover holes
14. Water pump mounting location
15. Idler shaft oil rifles.



01600282

Vee Cavity, QSK45

1. Water jacket header plate mounting capscrew holes
2. Oil supply to oil coolers
3. Water manifold holes used in manufacturing of block
4. Oil return from oil coolers
5. Idler shaft oil rifle
6. Thermostat housing mounting capscrew holes
7. Coolant from oil cooler cavity
8. Water flow from cooling passages around liners to thermostat housing.

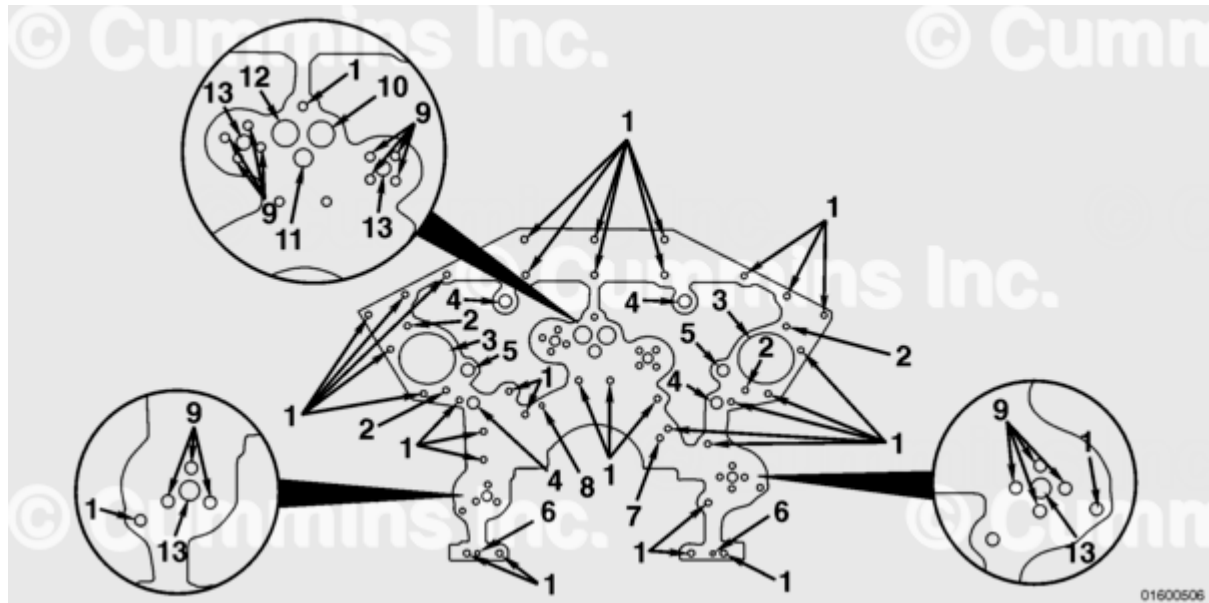


00600247

Vee Cavity, QSK60

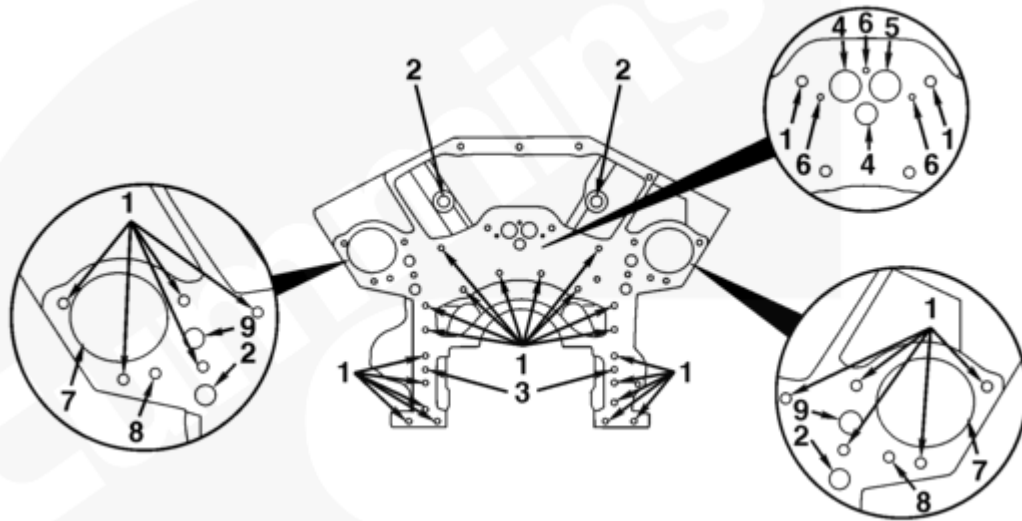
1. Water jacket header plate mounting capscrew holes
2. Oil supply to oil coolers
3. Water manifold holes used in manufacturing of block
4. Oil return from oil coolers

5. Idler shaft oil rifle
6. Thermostat housing mounting capscrew holes
7. Coolant from oil cooler cavity
8. Water flow from cooling passages around liners to thermostat housing.



Cylinder Block (Front)

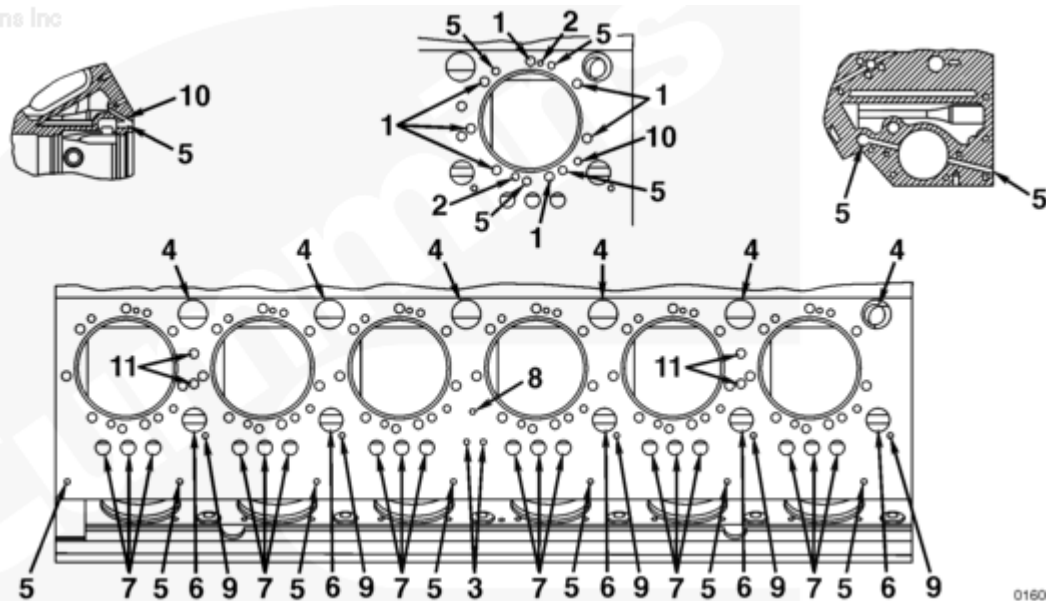
1. Front gear housing mounting capscrew holes (M16 x 2.5)
2. Camshaft thrust washer mounting capscrew holes (M10 x 1.5-6)
3. Camshaft bore (112 ± 0.025 mm [4.409 ± 0.001 in])
4. Coolant passages (28.56 ± 0.038 mm [1.124 ± 0.001 in])
5. Camshaft oil passages (28.56 ± 0.038 mm [1.124 ± 0.001 in])
6. Front cover dowel hole (11.964 mm [0.471 in])
7. Idler oil rifle (10 mm [0.394 in])
8. Idler oil rifle (10 mm [0.394 in])
9. Idler gear retaining capscrew holes (M16 x 2.16)
10. Main oil rifle from oil coolers (41.5 ± 0.038 mm [1.634 ± 0.001 in])
11. Main oil rifle supply
12. Main oil rifle to oil coolers (41.5 ± 0.038 mm [1.634 ± 0.001 in])
13. Idler gear shaft bore (22 ± 0.025 mm [0.866 ± 0.001 in]).



01600284

Cylinder Block (Rear)

1. Flywheel housing mounting capscrew holes (M16 x 2.6 and M10 x 1.5-6)
2. Coolant passages (26 mm [1.024 in])
3. Flywheel alignment dowel holes
4. Main oil rifle to oil cooler (38 mm [1.496 in])
5. Main rifle from oil cooler (38 mm [1.496 in])
6. Oil rifle blanking plate capscrew holes (M6 x 1.5)
7. Camshaft bore (112 ± 0.025 mm [4.409 ± 0.001 in])
8. Oil return drilling (8 mm [0.315 in])
9. Piston cooling nozzle rifles.



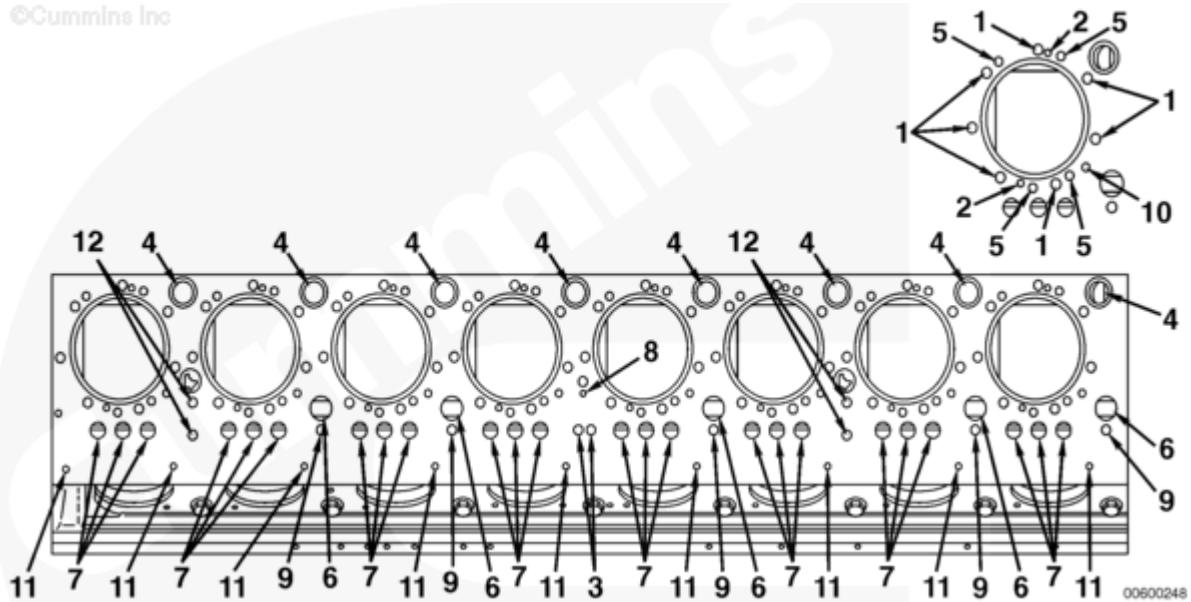
01600250

Cylinder Head Face, QSK45

1. Cylinder head retaining capscrew holes (M16 x 1.5 - 4H5H)
2. Cylinder head locating dowel holes
3. Fuel passages

4. Water manifold holes used in manufacturing of block
5. Coolant passages
6. Turbocharger oil drain holes
7. Push rod holes
8. Camshaft oil rifle
9. Turbocharger oil drain mounting capscrew holes (M10 x 1.5)
10. Oil passages to overhead
11. Lifting bracket mounting holes.

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Cylinder Head Face, QSK60

1. Cylinder head retaining capscrew holes (M6 x 1.5 - 4H5H)
2. Cylinder head locating dowel holes
3. Fuel passages
4. Water manifold holes used in manufacturing block
5. Coolant passages
6. Turbocharger oil drain holes
7. Push rod holes
8. Camshaft oil rifle
9. Turbocharger oil drain mounting capscrew holes (M10 x 1.5)
10. Oil passages to overhead
11. Lifting bracket mounting capscrew holes
12. Cam follower oil feed drillings.

Preparatory Steps

- Remove the engine. Refer to Procedure 000-001 in Section 0. (/qs3/pubsys2/xml/en/procedures/84/84-000-

001.html)

- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/84/84-002-004-tr.html)
- Remove the gear housing. Reference the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 001-033 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-033-tr.html) Reference the following procedure in the QSK45G and QSK60G Industrial Series Engine, Troubleshooting and Repair Manual, Bulletin 4021383, and the QSK60G Generator CM700 Troubleshooting and Repair Manual, Bulletin 4021396. Refer to Procedure 001-033 in Section 1. (/qs3/pubsys2/xml/en/procedures/84/84-001-033-tr.html)
- Remove the crankshaft. Refer to Procedure 001-016 in Section 1. (/qs3/pubsys2/xml/en/procedures/84/84-001-016.html)
- Remove the cylinder liners. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/84/84-001-028-tr.html)
- Inspect or remove the camshaft bushing. Refer to Procedure 001-010 in Section 1. (/qs3/pubsys2/xml/en/procedures/84/84-001-010-tr.html)



ck800wa

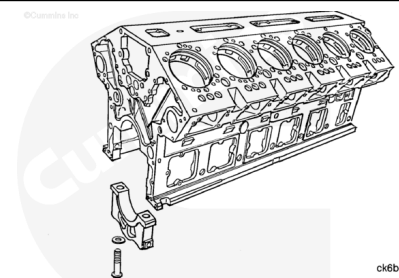
Clean and Inspect for Reuse

⚠ CAUTION ⚠

Use care when cleaning the cylinder block. Machined gasket surfaces and camshaft bushings can be damaged.

Use a scraper, wire brush, or Scotch-Brite™ pad to clean the heavy dirt deposits from the cylinder block:

- Gasket surfaces
- Mounting surfaces
- Cylinder liner counterbore ledge and the press fit areas
- Cylinder liner packing ring bore
- Top of the cylinder block
- Main bearing saddles and caps
- Cup plug bores.

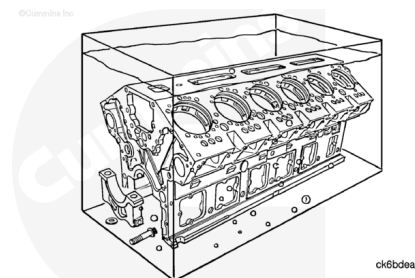


ck8bdwa

Use a bottle brush with a long handle to clean all of the oil passages.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Make sure the cleaning solution will not damage the camshaft bushings or the AVSEAL® sleeves.

Remove the cylinder block from the engine stand and place it in a cleaning tank.

Follow the instructions of the manufacturer of the cleaning tank and the manufacturer of the cleaning solution.

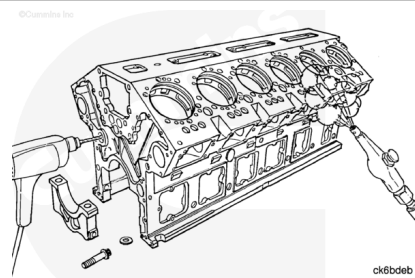
Cummins Inc. does **not** recommend any specific cleaning solution.

The best results are obtained using a cleaning solution which can be heated to 82 to 93°C [180 to 200°F].

Cleaning tanks that filter and mix the cleaning solution produce the best results.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

To reduce the possibility of cylinder block damage, make sure all of the water is removed from the capscrew holes and oil passages.

Remove the cylinder block from the cleaning tank.

Clean the cylinder block with a steam cleaner. Make sure all of the oil passages are clean.

Dry the cylinder block with compressed air.

If the cylinder block is **not** going to be reused immediately, apply a coating of preservative oil to prevent the formation of rust.

Cover the cylinder block to prevent dirt from sticking to the oil.

Inspect

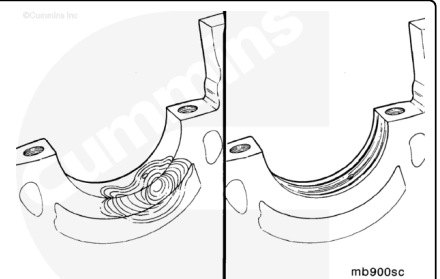
Main Bearing Bore

Inspect the main bearing bore saddle area for discoloration or damage due to high temperature.

Discoloration from high temperature indicates the need for a thorough inspection. Repair the main bearing bore, if necessary. Reference the Machine section in this procedure.

Check the minimum hardness of the cylinder block in the heat affected area.

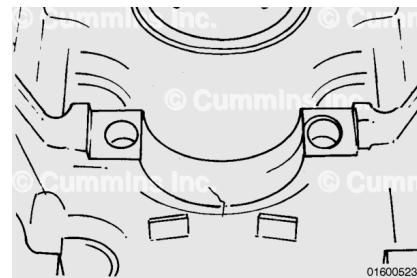
Minimum hardness of the cylinder block should be 179HB or above.



Note : When conducting a hardness test, be sure to use a maximum of five dents per test area, as an excessive number of dents will result in irreparable block damage.

Use crack detection kit, Part Number 3375432, or equivalent, to check the main bore saddle area for cracks.

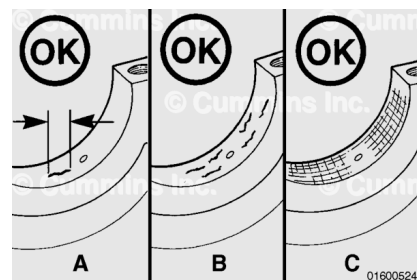
If cracks are found, follow the guidelines below.



The guidelines for acceptable cracks are as follows.

- Circumferential cracks that are no longer than 19.050 mm [3/4 in]
- Up to 6 cracks 9.525 mm [3/8 in] and shorter in the saddle area of the bore
- Porosity pockets less than 2.28 mm [0.090 in] diameter.

If any of the main bearing bores contain any of these criteria, the bearing bore **must** be repaired. Reference the repair procedure in the Machine section in this procedure.

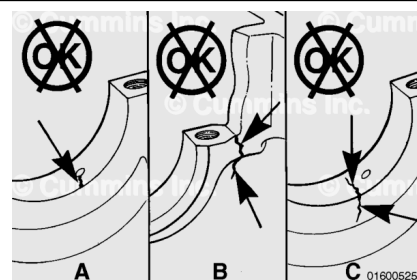


The guidelines for unacceptable cracks are as follows:

- Cracks entering the oil hole
- Cracks in the radii of the structural webs
- Cracks that break over the side of the main bearing bore saddle.

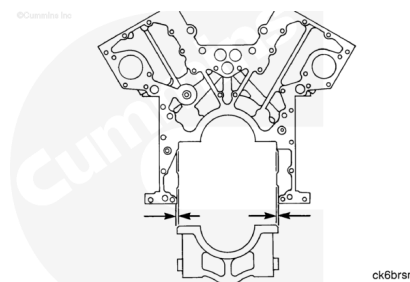
If any main bearing bores contain **any** of these cracks, the block can **not** be repaired and **must** be replaced.

Note : If a block has suffered from a spun bearing that has resulted in excessive heat in the main bearing saddle in the location of the internal fuel drillings, it is recommended to **not** reuse the block. This is due to the susceptibility of a crack at this location that can not be identified using normal test methods.



Check the main bearing caps that are loose. The main bearing cap **must** be replaced if it is **not** 0.000 to 0.25 mm [0.000 to 0.01 in] larger than the block.

New main bearing caps do **not** have the bore machined to a final specification. If a main bearing cap is replaced, the main bearing bore **must** be machined. Reference the Machine section in this procedure.



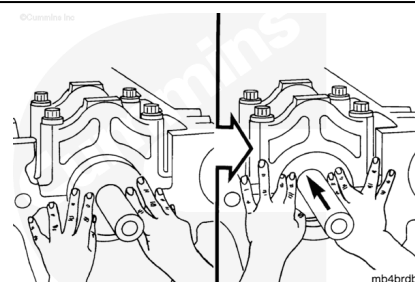
Install the main bearing caps and tighten to specification.

Refer to Procedure 001-016 in Section 1.

(/qs3/pubsys2/xml/en/procedures/84/84-001-016.html)

Slide the checking ring into the main bearing bore. If the checking ring does **not** slide through the bore, check the bore for burrs.

If the checking ring still does **not** slide through the bore, the bore is undersize and **must** be repaired.



Part Description	Part Number
Centering Ring	3164873
Checking Ring	3164872
Checking Bar	3375061

Use a 0.076 mm [0.003 in] feeler gauge that is a maximum of 13 mm [0.5 in] wide.

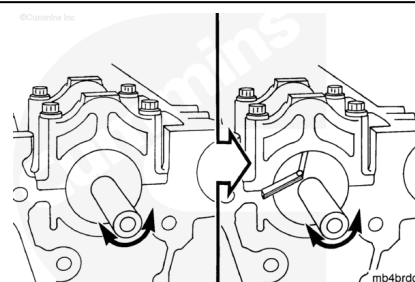
Center the checking ring in the bore. Attempt to put the feeler gauge between the checking ring and the bore. Rotate the feeler gauge in the bore at both sides of the checking ring.

The main bearing bore alignment is acceptable if:

- The gauge does **not** enter at any point
- The gauge will enter, but will **not** slide through or around the bore, and the alignment bar will rotate with the gauge inserted.

The main bearing bore alignment is **not** acceptable if:

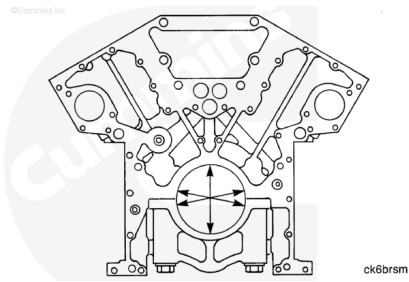
- The gauge enters and slides around the bore. (The bore is oversize and **must** be machined.)
- The gauge will enter on one side **only**, but can slide around the bore. (The bore is tapered and **must** be machined.)



If the tools to check the main bearing bore alignment are **not** available, use a bore dial indicator.

Measure the inside diameter in the three positions indicated in the illustration.

Make sure the weight of the bore dial indicator does **not** interfere with the measurement. Keep the solid stem of the indicator against the wall of the bore.



Main Bearing Bore Inside Diameter (Capscrews
Tightened to Specifications)

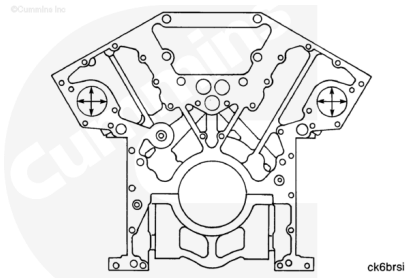
mm		in
192.901	MIN	7.594
192.951	MAX	7.596

The inside diameter of each bore **must** be completely round within 0.013 mm [0.0005 in].

If the main bearing bore alignment is **not** within specifications, the main bearing bore **must** be machined. Reference the Machine section in this procedure.

Camshaft Bore

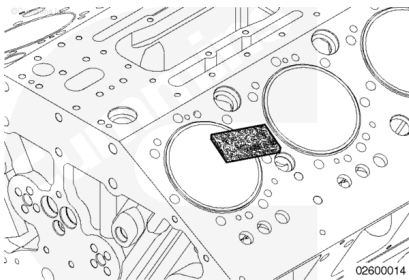
Inspect the camshaft bushing and bores for reuse. Refer to Procedure 001-010 in Section 1.
(/qs3/pubsys2/xml/en/procedures/84/84-001-010-tr.html)



Head Deck Surface

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use a scraper or a fibrous, abrasive pad, such as Scotch-Brite™ 7477, or equivalent, and solvent. Clean the top deck surface of the block. Do **not** allow **any** dirt into the cylinder. The surface

must be free of dirt, oil, and gasket material, but does **not** have to appear like new metal.

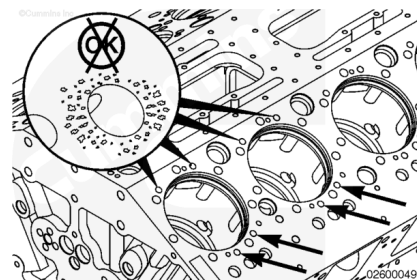
Check the top surface of the block for wear. If fretting damage is present in the area where a head gasket seal ring or a grommet makes contact, the surface **must** be repaired.

Reference the Machine section in this procedure.

Fretting damage in any other area is acceptable **only** if it does **not** change the protrusion measurement of the counterbore or liner.

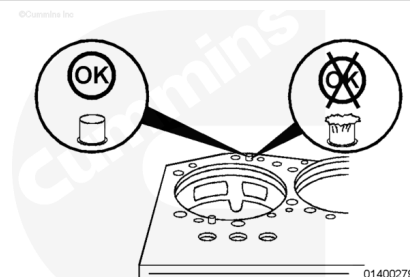
Fretting or pitting of the cylinder block head deck is **not** uncommon. It is **only** damaging to the head gasket joint if the fretting or pitting is close to the coolant or oil passages. These areas, if damaged, prevent the head gasket from sealing.

If the block is damaged, the cylinder block head deck **must** be machined. Reference the Machine section in this procedure.



Inspect the dowel pins for damage.

Use the dowel pin extractor, Part Number ST-1134, or equivalent, to remove the pins if they are damaged.



Use a precision straight edge (17) to check the head deck of the cylinder block for wear.

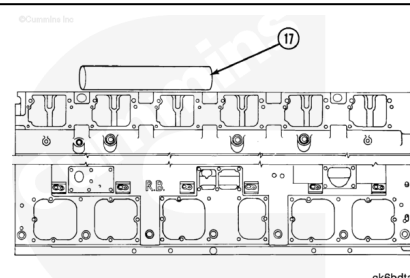
If fretting is visible in an area where a head gasket seal ring or a grommet touches the surface, the surface **must** be repaired.

Fretting damage in any other area is acceptable if it does **not** reduce the ability to measure the liner protrusion accurately.

Newly machined surfaces **must** be flat within 0.51 mm [0.020 in] under a cylinder head.

Waves on the surface are acceptable as long as they are **not** more than 0.018 mm [0.0007 in] in height. The high and low points of the waves **must not** be any closer than 25.4 mm [1.0 in].

If the top surface of the block is machined, the ledge depth of the cylinder liner counterbore **must** be machined.

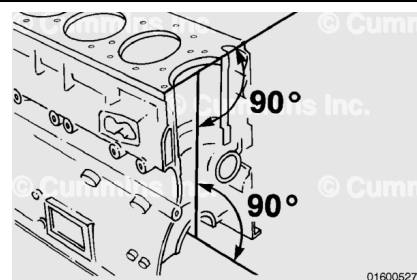


Check the top deck surface to make sure it is parallel and perpendicular to the main bearing bore.

If it is **not**, the top deck will require resurfacing.

Reference the Machine section for specifications.

Any main bearing bore repairs **must** be completed before resurfacing the top deck surface.

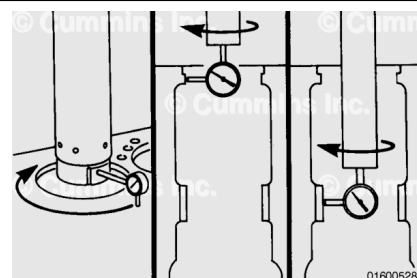


Check the concentricity reading of the cylinder bores. Refer to Procedure 001-027 in Section 1.

(</qs3/pubsys2/xml/en/procedures/84/84-001-027.html>)

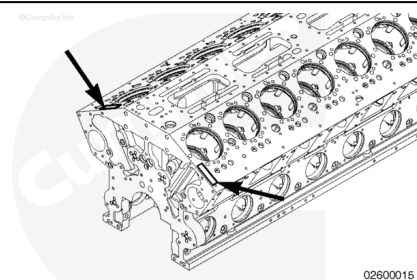
Compare against the flatness of the top deck surface.

A comparison of these values at 90-degree increments indicates the amount the bores and top deck surface are out of position with each other.



An oversize head gasket is often required on blocks that have had material removed from the top surface of the block.

Check the block for markings which indicate an oversize gasket is required due to prior salvage machining operations. These markings are often on the front left of the cylinder block head deck surface.

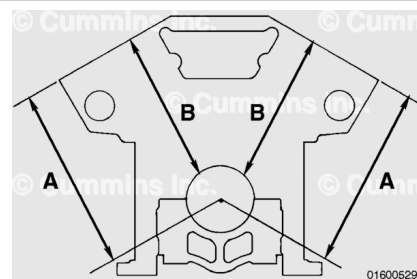


Inspect the cylinder block to determine which head gasket size is required.

The parting line of the main bearing cap is **not** the same as the center line of the main bearing bore.

Measure the cylinder block height.

Standard Block Height:



From crankshaft center line (A):

mm		in
529.525	MIN	20.8474
529.675	MAX	20.8533

From crankshaft bore (B):

mm		in
433.062	MIN	17.0497

From crankshaft bore (B):		
mm		in
433.212	MAX	17.0556

Block Deck Height from Crankshaft Center Line (A) for
Oversize Head Gaskets:

For +0.25 mm [0.010 in] oversize gasket:		
mm		in
529.275	MIN	20.8376
529.425	MAX	20.8435

For +0.50 mm [0.020 in] oversize gasket:		
mm		in
529.025	MIN	20.8278
529.175	MAX	20.8337

For +0.76 mm [0.030 in] oversize gasket:		
mm		in
528.765	MIN	20.8175
528.915	MAX	20.8234

Block Height from Crankshaft Bore (B) for Oversize Head
Gaskets:

For +0.25 mm [0.010 in] oversize gasket:		
mm		in
432.812	MIN	17.0398
432.962	MAX	17.0457

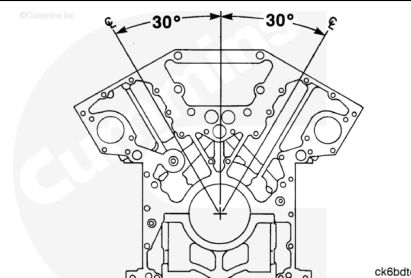
For +0.50 mm [0.020 in] oversize gasket:		
mm		in
432.562	MIN	17.0300
432.712	MAX	17.0359

For +0.76 mm [0.030 in] oversize gasket:		
mm		in
432.302	MIN	17.0198
432.452	MAX	17.0257

The height of the cylinder block **must not** vary more than 0.08 mm [0.003 in] from end to end of the cylinder block. If the cylinder block height is **not** within specifications, the top surface of the cylinder block **must** be machined or the block **must** be replaced.

If the head deck surface of the block is machined, the ledge depth of the cylinder liner counterbore **must not** vary more than 0.13 mm [0.0005 in] between the banks.

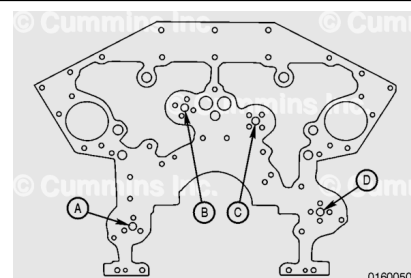
The cylinder bore angle does **not** need to be measured, but **must** be maintained at 30 degrees if the cylinder head surface is machined.



Front Gear Train

Measure the inside diameter of the bore for all four idler shafts.

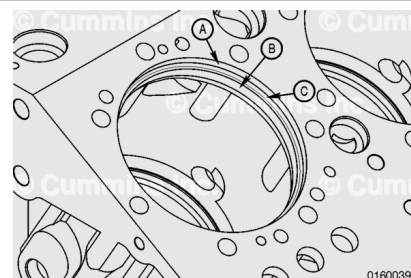
Idler Shaft Bore Internal Diameter		
mm		in
21.975	MIN	0.865
22.025	MAX	0.867



There is **no** Cummins Inc. approved salvage to repair the idler shaft bore.

Counterbore

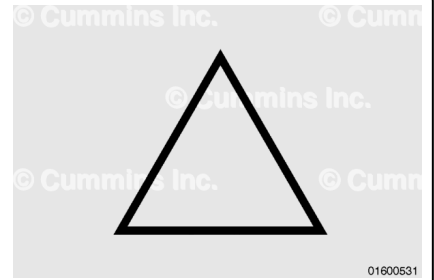
Inspect the upper cylinder counter bore and lower packing ring bore for reuse. Refer to Procedure 001-027 in Section 1. (</qs3/pubsys2/xml/en/procedures/84/84-001-027.html>)



Machine

⚠ CAUTION ⚠

Only a competent machinist should attempt these repairs with a fully equipped machine shop and/or using the service tools described in each process. The cylinder block could be irreversibly damaged and therefore not be reused.

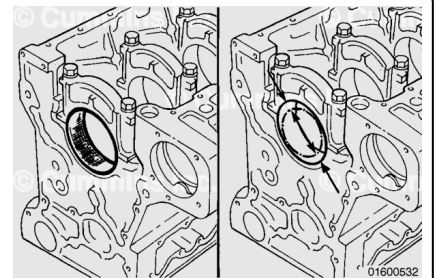


It is important that each procedure is read before beginning **any** repairs.

Make sure all required tools and parts are available before beginning repairs.

Main Bearing Bore

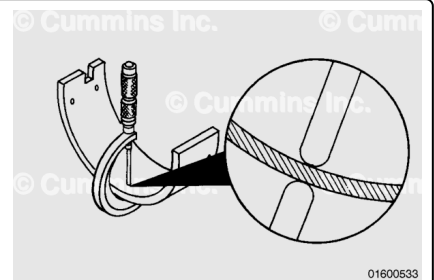
To repair the main bearing bore, a repair sleeve **must** be inserted into the bore to bring the bore back to original specifications. The main bearing bore **must** be machined larger to accept the repair sleeve.



Select the correct repair sleeve from the main bearing saddle inserts table, Part Number 3375168. Consult the Service Products Catalog, Bulletin 3377710.

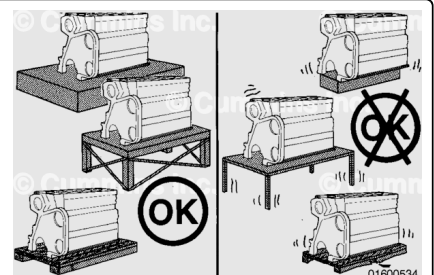
Measure the thickness of the repair sleeve. The nominal thickness of a Cummins® service repair sleeve is 2.54 mm [0.100 in]. The sleeve should measure within ± 0.25 mm [± 0.010 in] of this nominal thickness.

If the thickness of the sleeve is greater than this tolerance, the bore dimensions described in this procedure **do not** apply. Specifications for thicker sleeves are **not** addressed in this procedure.



⚠ WARNING ⚠

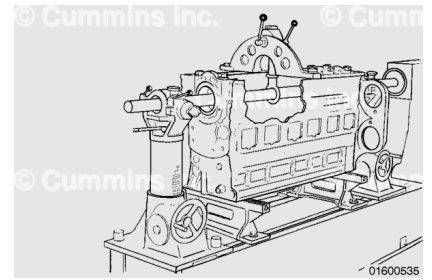
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Put the cylinder block on an adequately sized vibration-resistant work surface.

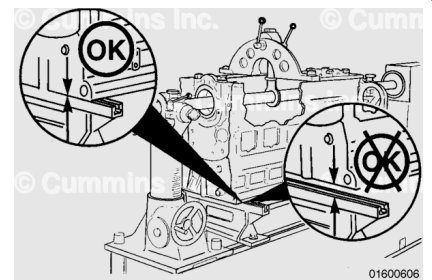
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

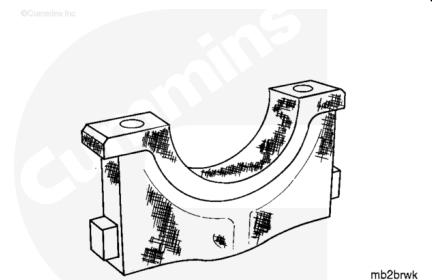


Position the cylinder block to permit the use of a steady rest during the machining operation.

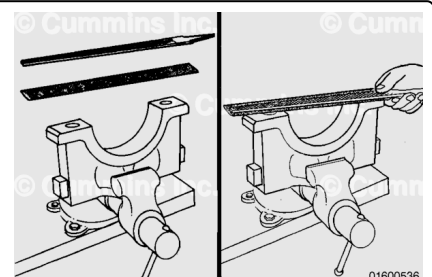
The cylinder block **must** sit squarely (flat) on the vibration-resistant work surface.



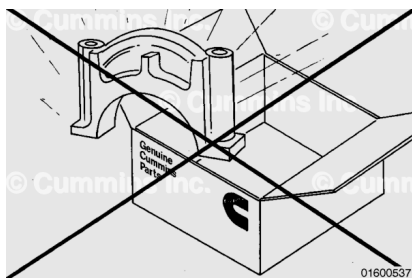
Select the damaged main bearing cap that was removed from the engine or another unserviceable main bearing cap that is appropriate for the engine and the main bearing location being repaired.



Remove any burrs from the joint face of the main bearing cap. A flat file wrapped in an emery cloth is effective.



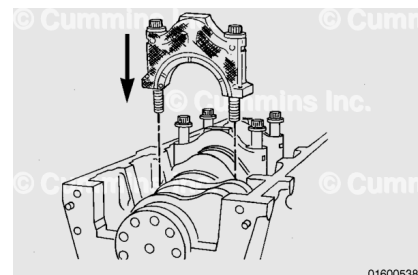
The main bearing cap will be destroyed during the cutting operation. Use a scrap or damaged cap. **Do not** install a new cap at this time.



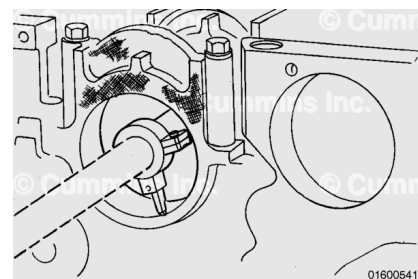
Install the unserviceable main bearing cap appropriate for the bore to be repaired.

Tighten the capscrews to specification. Refer to Procedure 001-016 in Section 1.

(/qs3/pubsys2/xml/en/procedures/84/84-001-016.html)

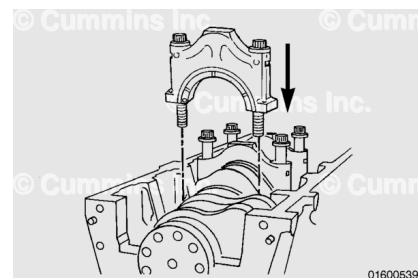


The main bearing cap can be reused on other engines while making a saddle repair. It should be retained as part of the process tooling.



Install the remaining main bearing caps. Refer to Procedure 001-016 in Section 1.

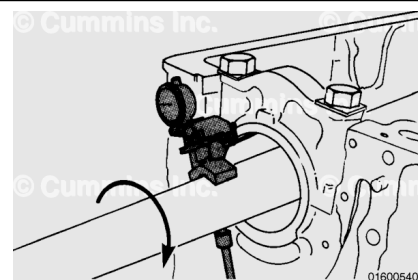
(/qs3/pubsys2/xml/en/procedures/84/84-001-016.html)



Use a dial indicator to align the center line of the main bearing bore with the center line of the boring bar.

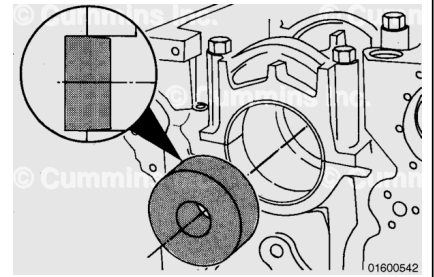
The most accurate alignment is achieved from the undamaged main bearing bores that are at the ends of the block.

Correct alignment of the bores is ± 0.025 mm [± 0.001 in].



Cummins Inc. recommends, for portable main bearing bore machining, the following tools are used for QK45/60 engines.

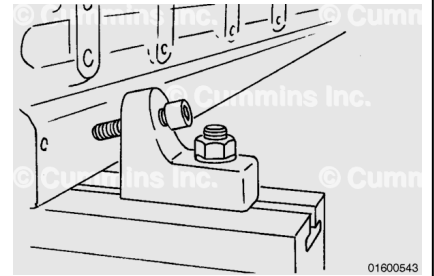
Part Description	Part Number
Mounting Adapter	3164879
Cutter Bits	3165027 and 3165028
Main Bearing Alignment Kit	3164874
Checking Bar/Boring Bar	3375061



Reference the specific tool instructions for tool usage.

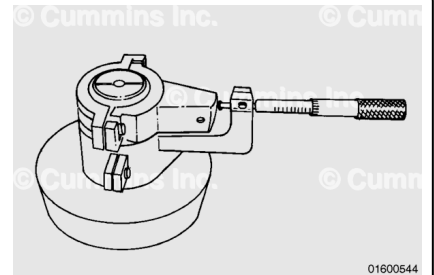
Clamp the cylinder block to the work table after correct alignment is achieved.

The cylinder block should **not** be permitted to move during the cutting operation.

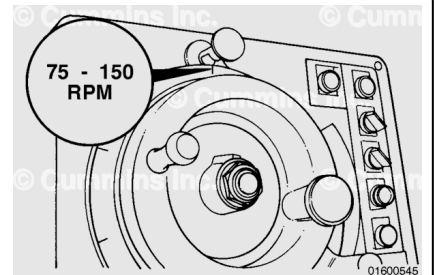


Adjust the boring bar cutter to remove 0.25 to 0.51 mm [0.010 to 0.020 in] of material from the inside diameter of the damaged main bearing bore.

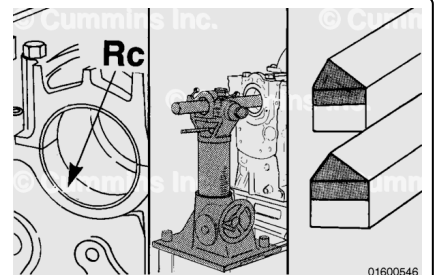
Reference the manufacturer's instructions for cutter depth adjustment procedures.



Operate the cutter at 75 to 150 rpm. If the current tool tip wears quickly or breaks, reduce the speed of the cutter.



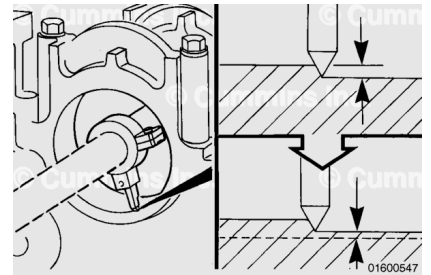
The amount of material that can be removed depends on the hardness of the surface material, type of boring machine, and the type of cutting tool being used.



Machine the bore to the following specifications.

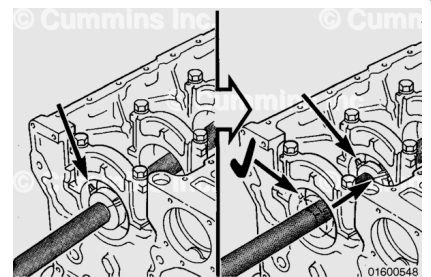
Enlarged Main Bearing Bore Ready for Repair Sleeve Insert

mm		in
195.40	MIN	7.693
195.50	MAX	7.697



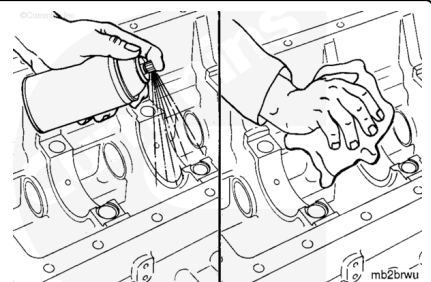
The final surface finish maximum specification is 2.03 μm [80 μin].

Complete the cutting operation on each main bearing bore before proceeding to the adjacent bore(s). This lessens the potential for misalignment between adjacent main bearing bores.

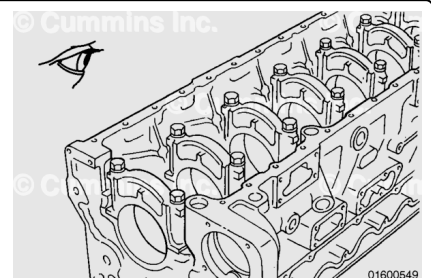


Clean the main bearing bore(s) to remove cutting oils and metallic fines.

Spray cleaner, Part Number 3375433, or equivalent can be used.



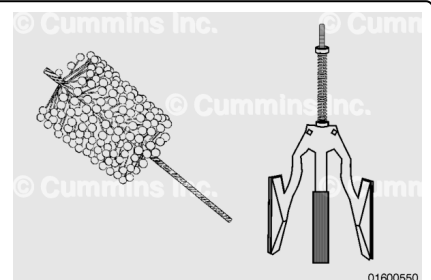
Inspect the enlarged main bearing bore for any fault indication that was **not** discovered during the initial inspection. This is especially critical in instances where the bore was distressed by heat or impact loading.



After the cutting operation is complete, use a rigid or flexible hone to remove any burrs from each of the main bearing bores.

Remove all burrs from the oil supply hole drillings.

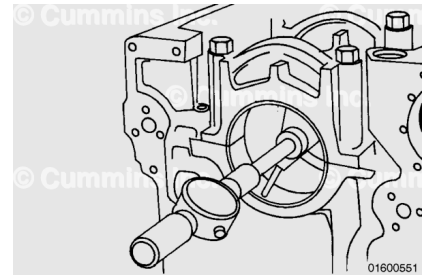
The surface finish **must** remain within the specification shown above.



Check the enlarged bearing bore(s) to be sure they are the correct size for the repair sleeve.

Measure each in four equally spaced directions.

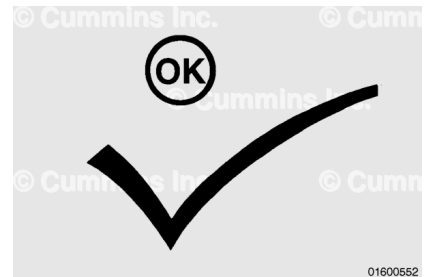
The diameter of the enlarged main bearing bore(s) should **not** vary more than ± 0.05 mm [± 0.002 in].



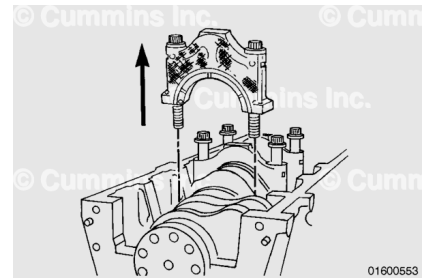
If the bore size is within specifications, proceed with the repair process.

If **not**, repair the main bore.

Consult the Alternative Repair Manual, Bulletin Number 3379035.



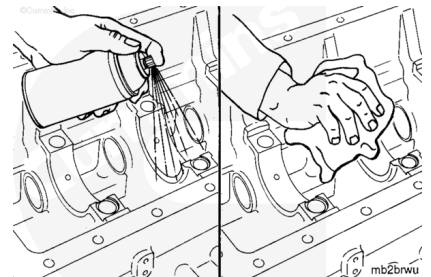
Remove the damaged main bearing cap(s) from the bore(s) that were cut for repair sleeves. Refer to Procedure 001-016 in Section 1. (</qs3/pubsys2/xml/en/procedures/84/84-001-016.html>)



Prepare the enlarged bore for installation of the repair sleeve.

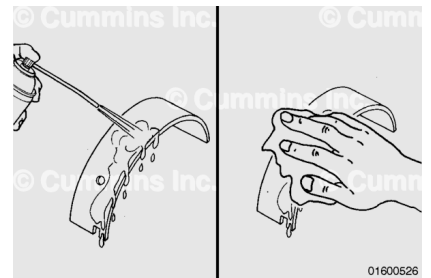
Clean the main bearing saddle area with spray cleaners, Part Numbers 3375433 and ST-1272-13, or equivalent.

Allow 2 minutes for the excess cleaner to evaporate.



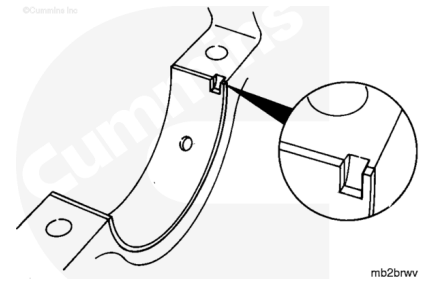
Clean the outside diameter of the repair sleeve.

Use spray cleaner, Part Numbers 3375433, ST-1272-13, or equivalent, to remove any oils, grease, or preservative coating.



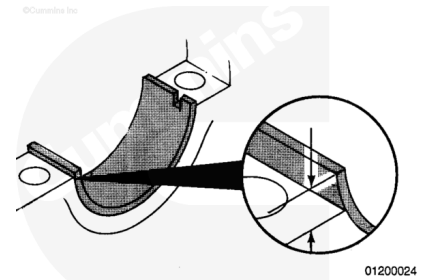
Align the main bearing locking tang slot in the repair sleeve with the locking tang slot in the cylinder block.

The oil hole in the repair sleeve **must** be centered over the oil hole in the block.



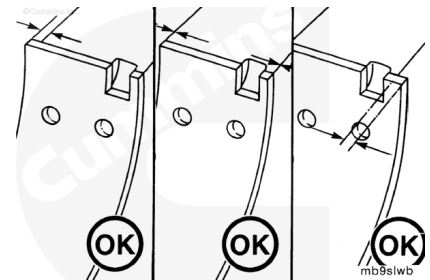
The ends of the repair sleeve should protrude above the main bearing bore joint faces.

In most cases, the protrusion will be approximately 1.52 mm [0.060 in] above the joint face of the cylinder block on each end of the repair sleeve.

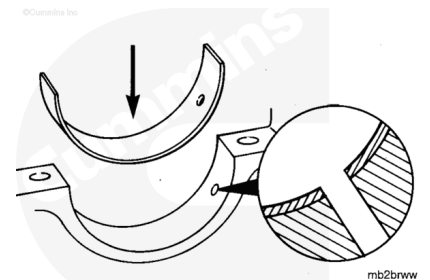


Check the side-to-side position of the repair sleeve. In most cases, the sleeve will be flush with the sides of the main bearing saddle.

A variance of ± 0.38 mm [± 0.015 in] from being flush with the saddle sidewall is acceptable.



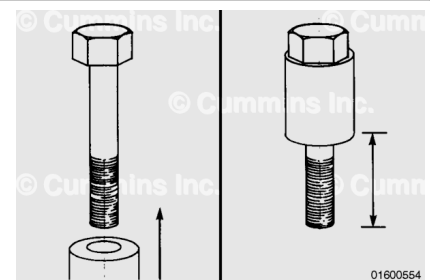
Check the location and position of the oil supply holes.



Select two main bearing cap retaining capscrews.

Install one of the spacers which is part of the main bearing saddle hardware kit on to each of these capscrews, so that approximately 86.2 mm [3 in] of the capscrew protrudes beyond the end of the spacer.

Consult the Service Products Catalog, Bulletin 3377710.

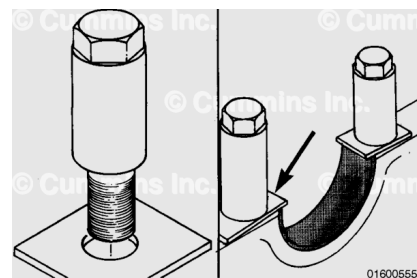


Install a flat washer or the locking plate on to each of the capscrew and spacer assemblies. These locking plates are part of the main bearing saddle hardware kit. Consult the

Service Products Catalog, Bulletin 3377710.

Install the capscrew assemblies in the cylinder block.

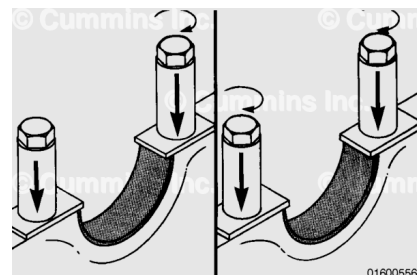
Position the locking washer so that the edge of the repair sleeve is held down by the flat washer or locking plate.



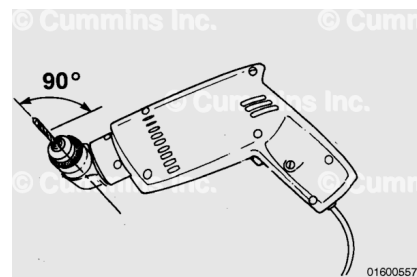
Tighten the capscrews and washer assemblies finger-tight.

Alternately tighten each of the capscrew assemblies.

Torque Value: 55 n•m [41 ft-lb]

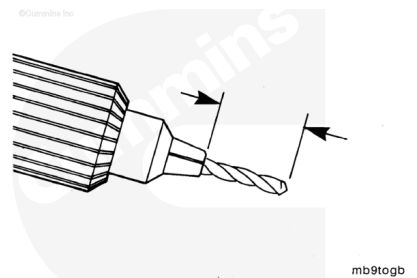


A 6 mm [1/4 in] capacity drill motor and a 90 degree (right angle) drill is required for most Vee design engines, due to the position of the retaining rivets.



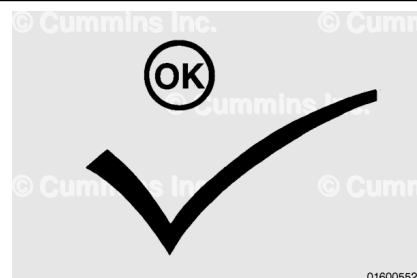
Install a 3.175 mm [1/8 in] diameter drill bit in the drill chuck.

Install a drill stop on the drill so the drilled depth is controlled to between 5.85 to 6.35 mm [0.230 to 0.250 in].



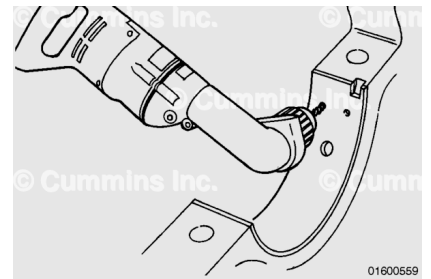
The correct drill depth setting is **very important** for making a successful repair.

If the drill depth is too shallow or too deep, the rivet will **not** hold effectively.

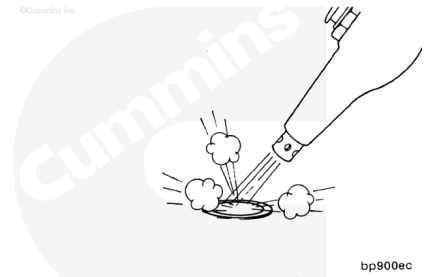


Drill the cylinder block to accept the retaining rivets. Use the holes in the repair sleeve to determine the location and quantity of holes required.

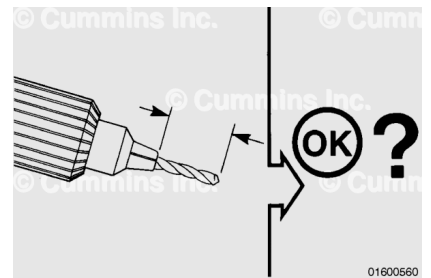
Be sure to drill all the required rivet holes.



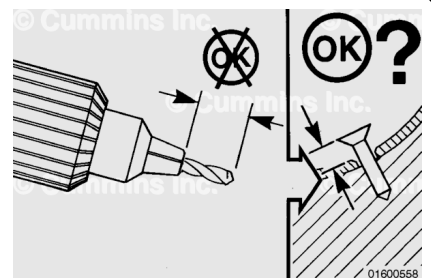
Evacuate shavings and other debris from the drilled holes.



Check the position of the drill stop to be sure the original depth setting is still correct.

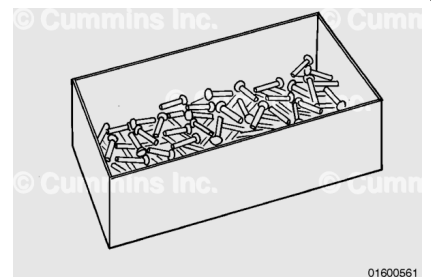


If the drill bit has moved, be sure the drilled holes will permit the rivet to protrude approximately 2.54 mm [0.100 in] above the surface of the repair sleeve.

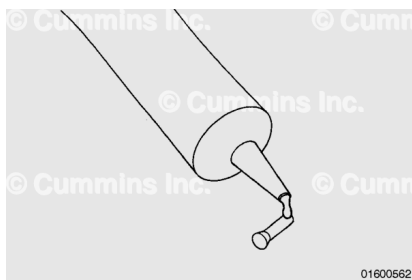


Select the correct quantity of retaining rivets.

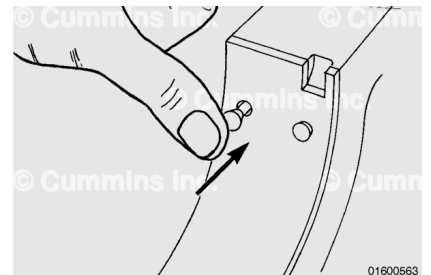
The rivets come as part of a kit with the salvage sleeves.
Consult the Service Products Catalog, Bulletin 3377710.



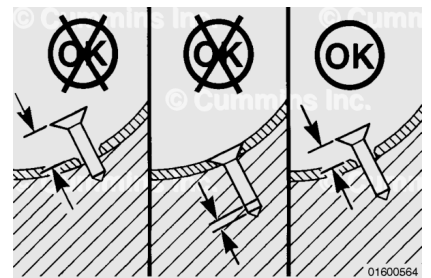
Put a small amount of sealant, Loctite™ 601, or equivalent, on the tip and body of the rivet.



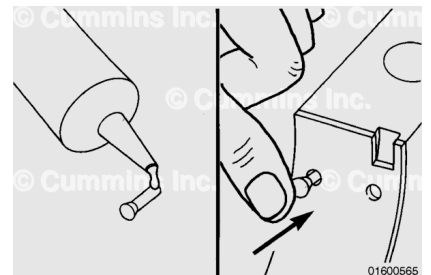
Push the rivet through one of the drilled holes in the repair sleeve and into the drilled hole in the cylinder block.



The rivet should protrude above the surface of the repair sleeve by approximately 2.54 mm [0.100 in] when correctly installed.

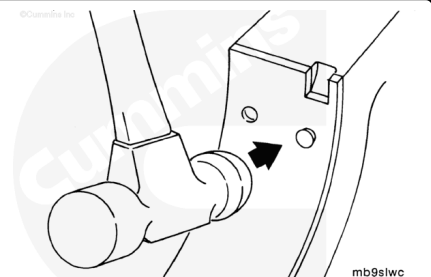


Prepare and install the remaining rivets.



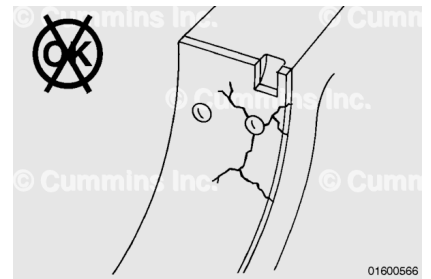
Peen the rivets to hold the repair sleeve in place.

Use the ball end of a 0.9 kg [2 lb] hammer (approximate weight and size) for the peening operation.



⚠ CAUTION ⚠

The repair sleeve can be fractured if hit during the peening process.

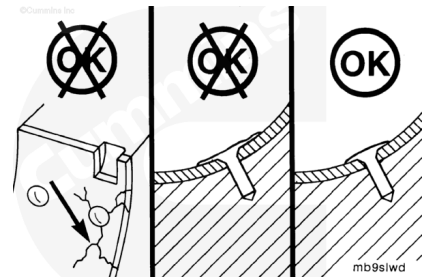


Do **not** use **any** fractured repair sleeves.

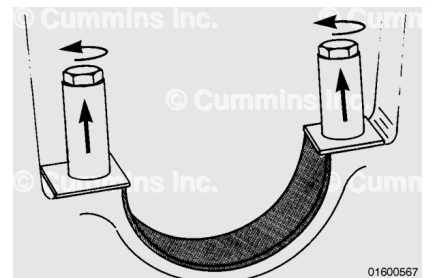
The retaining rivets **must** be fully peened. Do **not** peen excessively.

Excessive peening will result in rupturing or splitting of the rivet head.

A uniform peening pattern provides the best results.

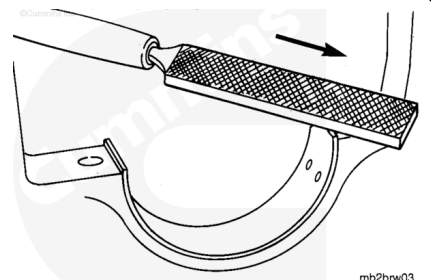


Remove the retaining capscrews, spacer, and washer assemblies.



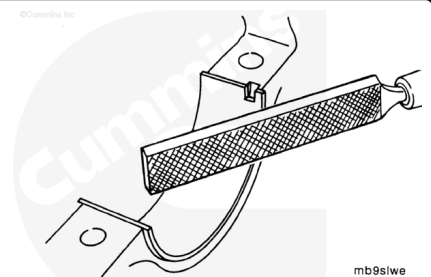
Cut the ends and sides of the repair sleeve flush with the main bearing cap joint face of the cylinder block and the sides of the main bearing bore saddle.

A 203.2 mm [8 in] mill file can be used to perform this operation.



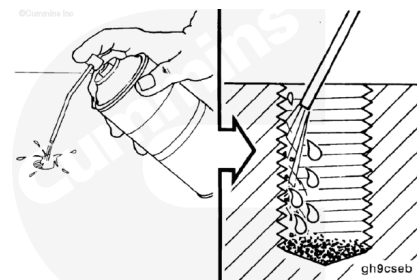
Cut the edge(s) of the repair sleeve flush with the sides of the main bearing bore saddle.

A 203.2 mm [8 in] mill file can be used to perform this operation.



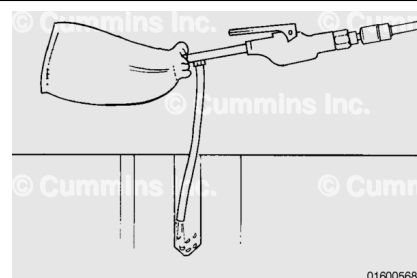
Clean the main bearing saddle and the main bearing capscrew threads.

Flush the threads with cleaner, Part Number 3375433, to remove filing fines.



Remove the filing fines and excess cleaner from the main bearing capscrew holes.

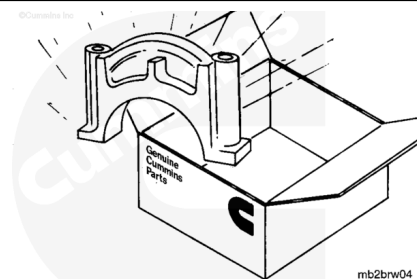
A chip removing unit, Part Number ST-1272-11, can be used for this operation.



Select the appropriate new replacement semi-finished main bearing cap or a good used main bearing cap.

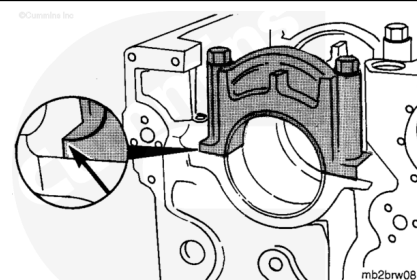
Be sure the cap is the correct one for the engine and main bearing location.

Reference the Cylinder Block option in the parts catalog for the engine to determine the correct main bearing cap part number.



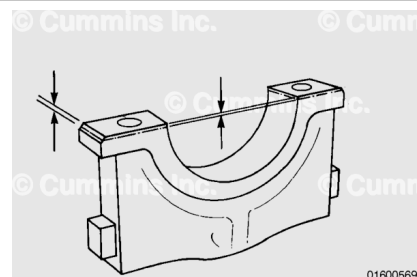
Check to be sure the replacement cap has the correct fit in the cylinder block press fit area(s).

Reference the Main Bearing Bore in the Inspect section in this procedure.



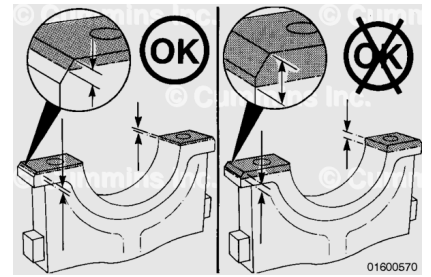
A used main bearing cap in good condition can be used.

The thickness of the parting face of the main bearing joint face **must** usually be reduced by 0.25 mm [0.010 in].



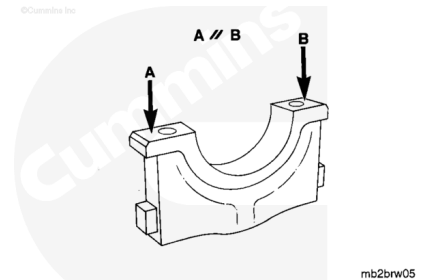
Do **not** cut excessive amounts from the joint face of the main bearing cap.

Excessive cutting will **not** allow installation of the main bearing side bolts and/or correct installation of the main bearing cap retaining screws.



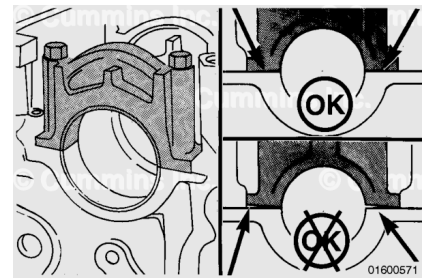
It is very important that the joint faces of the cap are cut parallel to each other and square to their mating surfaces in the cylinder block.

A high quality main bearing cap cutting machine **must** be used to cut the joint face of the cap.



The main bearing cap **must** fit squarely on the cylinder block's joint face.

Main bearing caps that do **not** fit correctly can change assembly clearances and tolerances.



Check the squareness of the mating surface of the main bearing cap.

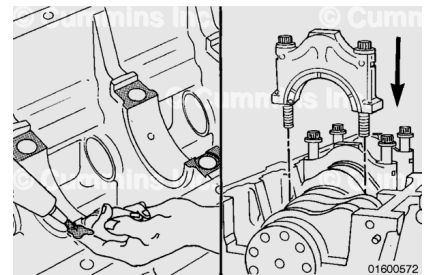
The joint faces of the cap **must** sit squarely with the cylinder block joint face, as determined by a bearing bluing compound.

Apply a light coating of bluing to the cylinder block joint face.

Install the main bearing cap.

Tighten the capscrews to approximately:

Torque Value: 135 n·m [100 ft-lb]

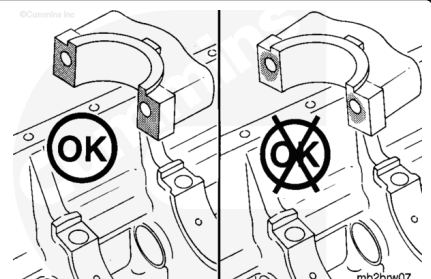


Remove the retaining capscrews and the main bearing cap. Refer to Procedure 001-016 in Section 1.

(/qs3/pubsys2/xml/en/procedures/84/84-001-016.html)

The bluing should have transferred to the joint face of the main bearing cap.

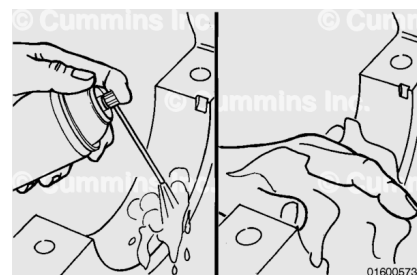
A bluing transfer of less than 80 percent of the area of the joint face(s) is unacceptable.



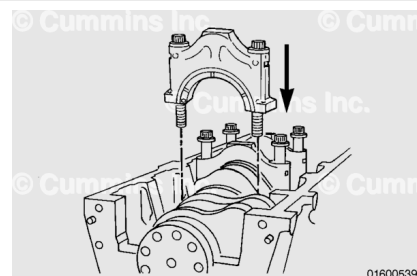
Only use main bearing caps that have a bluing transfer of 80 percent or greater.

Remove the bluing from the joint faces and main bearing bore of the main cylinder.

Use a soft cloth and spray cleaner, Part Number 3375433, or equivalent.



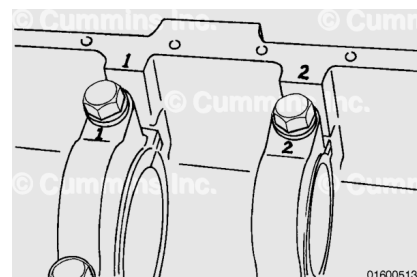
Install the new replacement main bearing cap. Use the following procedure for capscrew torques. Refer to Procedure 001-016 in Section 1.
(/qs3/pubsys2/xml/en/procedures/84/84-001-016.html)



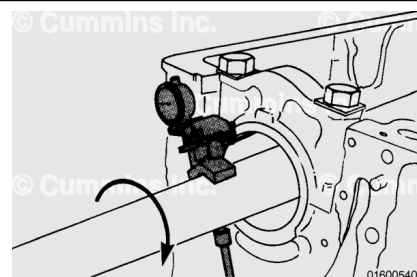
Mark the replacement cap to identify its location and position.

Steel stamps with 6.35 mm [1/4 in] characters are recommended.

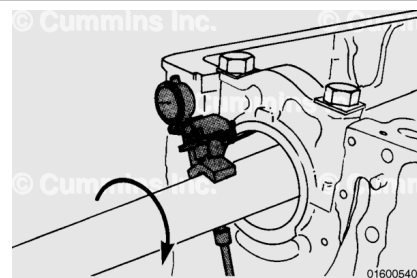
Be sure to mark the cap in the same area as the other main bearing caps.



Use a dial indicator to align the center line of the main bearing bore with the center line of the boring bore.

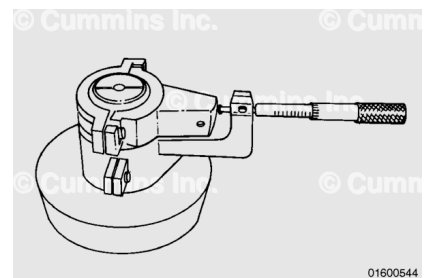


The most accurate alignment is achieved from the undamaged main bearing bores that are the furthest from the damaged bores. These are usually the bores that are at the ends of the cylinder block.



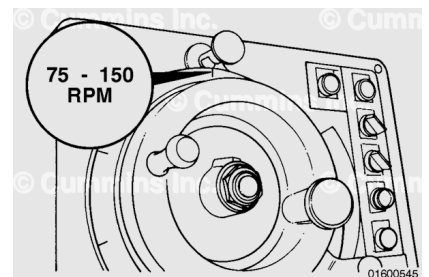
Adjust the boring bar cutter to remove 0.25 to 0.51 mm [0.010 to 0.020 in] of material from the inside diameter of the main bearing bore.

Reference the manufacturer's instructions for cutter depth adjustment procedures.



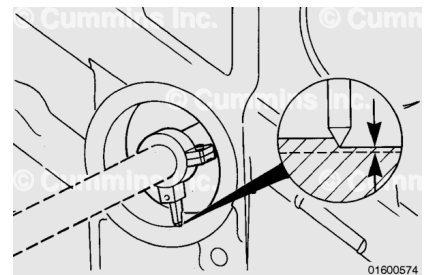
Operate the cutter at 75 to 150 rpm.

If the cutting tool tip wears quickly or breaks, reduce the speed of the cutter.

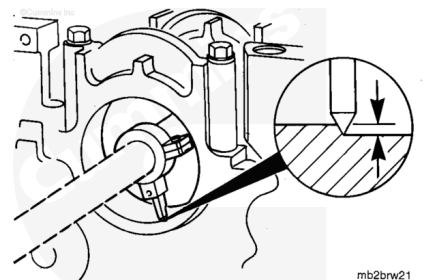


The amount of material that can be removed will depend on the following.

- Material hardness
- Amount the rivet protrudes from the surface of the repair sleeve
- Type of boring machine
- The surface finish requirements.

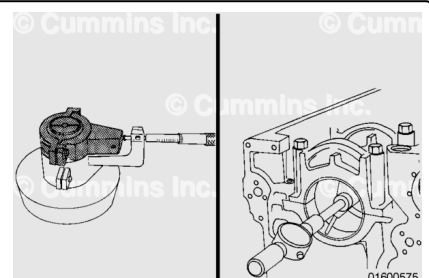


After cutting the main bearing bore the first time, approximately 1.52 mm [0.060 in] of material can be removed from the bore diameter with each additional cut.



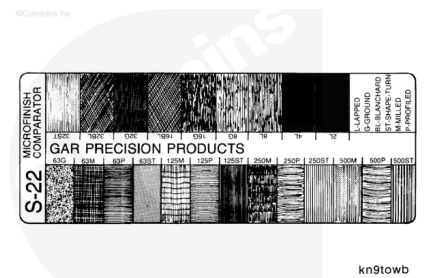
After making the second cut, check the accuracy of the cutting tool by comparing the cut diameter of the main bearing bore with the set diameter of the cutting tool.

This size comparison will reduce the chances of error in setting the cutting tool and allow the bore to be cut to the proper finished size.

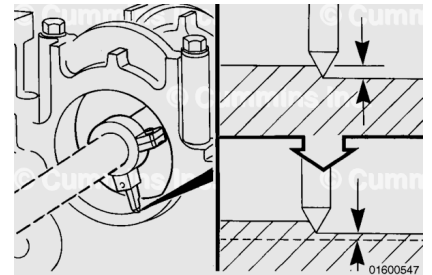


The surface finish **must** remain within the specification shown above.

Reduce the depth of the cut and reduce the cutter speed of feed if the surface finish is **not** acceptable.

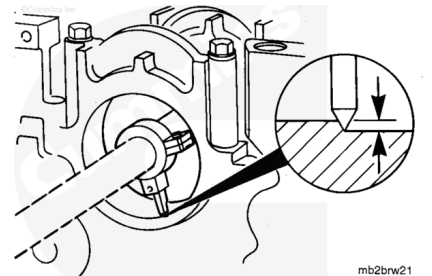


Continue to cut the inside diameter of the main bearing bore until it is 0.13 mm [0.005 in] smaller than the required size.

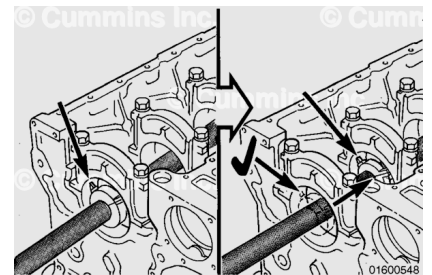


Remove the remaining 0.13 mm [0.005 in] of material from the main bearing bore. Use reduced depth of cutting operations.

Check the cut diameter to be sure it is within new bore specifications. Refer to the Main Bearing Bore information in the Inspect section of this procedure.



Complete the cutting operation on each main bearing bore before proceeding to adjacent bore(s). This lessens the potential for misalignment between adjacent bearing bores.

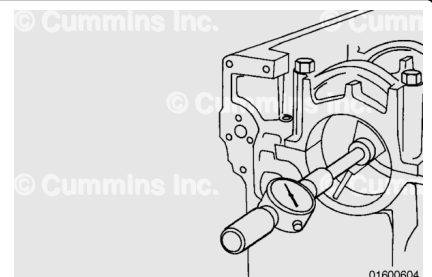


⚠ CAUTION ⚠

The main bearing bore must be within specification tolerances. Engine damage will result if bores are out of tolerance.

Measure the newly cut bore diameter. Be sure it is within the stated size tolerance. Refer to the Main Bearing Bore information in the Inspect section of this procedure.

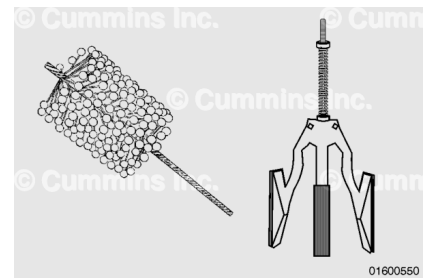
If the bore is too small, it **must** be machined to make the bore larger.



If the bore is too large, remove the repair sleeve and begin the main bearing saddle repair process again.

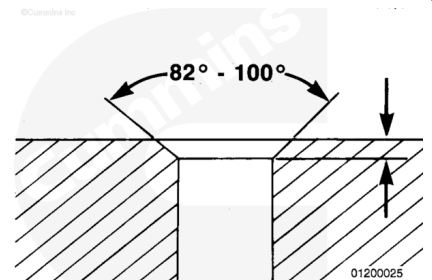
After the cutting operation is complete, use a rigid or flexible hone to remove any burrs from each main bearing bore.

The surface finish **must** remain within the specifications shown above.



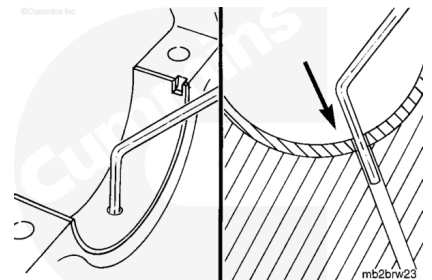
Chamfer the oil supply hole in the repair sleeve to remove any burrs.

The chamfer angle **must** be between 82 and 100 degrees included angles.



Make sure the oil supply holes in the cylinder block and main bearing bore repair sleeve are aligned.

If the holes are **not** aligned, the repair sleeve **must** be removed and the main bearing bore repair procedure **must** be started again.

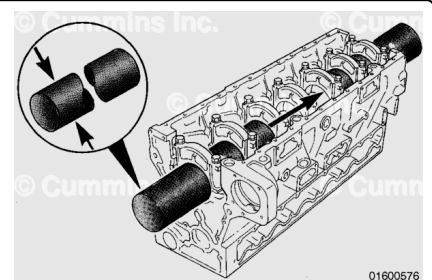


Check the alignment of the main bores for short bend alignment.

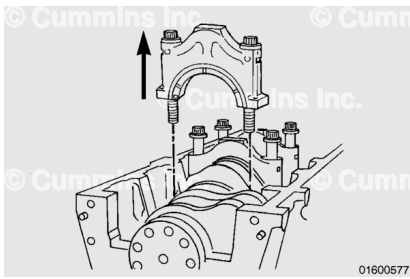
Short bend alignment is the alignment between adjacent main bearing bores.

Pass a solid bar 0.025 mm [0.001 in] smaller than the main bearing bore inside diameter through the bores of the cylinder block.

A main bearing alignment kit can also be used. Reference the service tool instructions for information on how to use the bearing alignment kit.

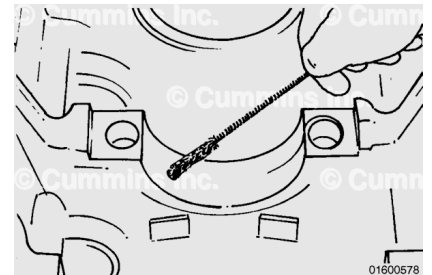


Remove the main bearing caps. Refer to Procedure 001-016 in Section 1. (</qs3/pubsys2/xml/en/procedures/84/84-001-016.html>)



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Brush the main bearing oil supply drillings to remove shavings and other debris.

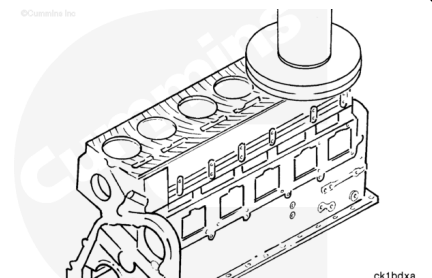
Use solvent and a nylon brush for this operation.

Head Deck Surface

Resurface the cylinder block, if required, after the main bearing bores have been machined, and before machining the counterbore ledge depth.

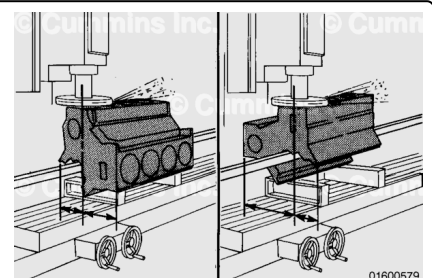
This machine process can **only** be done on a milling machine with a machine bed large enough to hold the entire cylinder block.

Do **not** remove too much material. This will result in piston to valve contact.



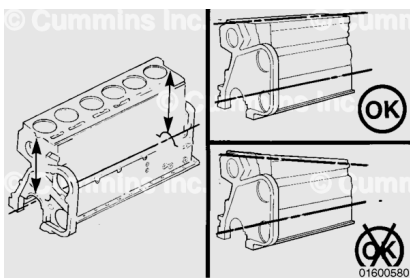
Determine how much material is to be removed before machining, to determine which oversize head gasket should be used. Reference the block deck height specifications above.

Machine the cylinder block head deck surface. Use the appropriate method for the milling machine being used to achieve the desired results.



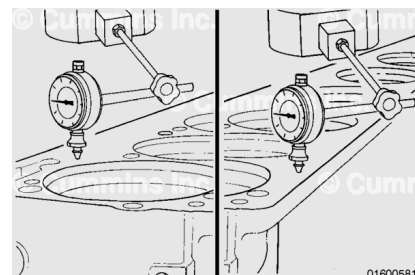
Check the new surface from end-to-end to determine the surface parallelism.

The surface should **not** vary more than 0.0375 mm [0.0015 in] for each 508 mm [20 in] of the cylinder block surface length.



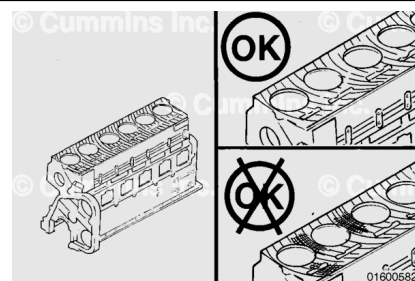
Check the new surface from side-to-side to determine surface perpendicularity.

The surface should **not** vary more than 0.02 mm [0.001 in] in 254 mm [10 in] of the cylinder block surface width.



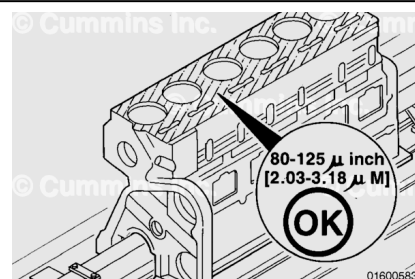
Check for discoloration, heat checks, or heat spots on the top deck surface. These deck conditions are **not** acceptable as a finished surface condition.

If the maximum amount of material has been removed and the block shows signs of discoloration and heat spots, check the cylinder block hardness at these positions. If the hardness is below 205HB, the block can **not** be reused.

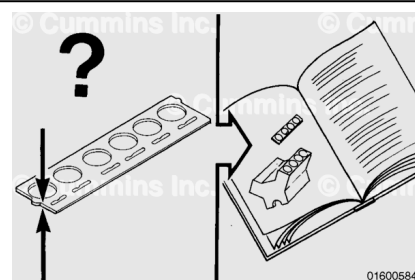


The following are the specifications for cylinder block head deck surface finish.

Cylinder Block Head Deck Surface Finish		
micro-m		micro-in
2.03	MIN	80
3.18	MAX	125



Reference the parts catalog to find information on the correct oversize head gasket to be used.

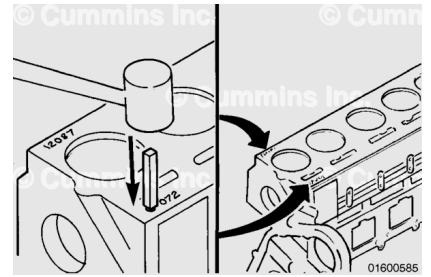


Mark the cylinder block to indicate the thickness of the head gasket that **must** be used with the machined cylinder block.

The preferred location to mark the cylinder block is on the left front top surface.

Use 1/4 inch characters. The common notation is STD.; 0.010, 0.020 for either the standard or oversize gaskets.

Be sure to remove any burrs caused by the stamping operation.

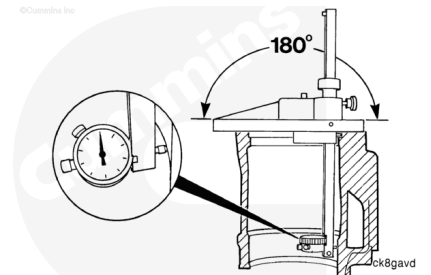


Measure the concentricity of each of the cylinder bores before you remove the cylinder block from the machine work surface.

Use the following procedure for cylinder bore concentricity specifications and machining processes. Refer to Procedure 001-027 in Section 1.

(/qs3/pubsys2/xml/en/procedures/84/84-001-027.html)

If any bores are outside of the specifications, mark them for repair.

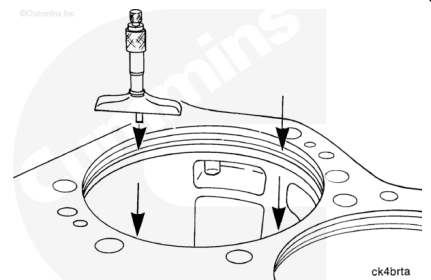


Measure each counterbore depth and compare with the specifications.

After the cylinder block head deck has been machined, all cylinder counterbores **must** be machined to the correct cylinder bore specifications.

Use the following procedure for cylinder bore concentricity specifications and machining processes. Refer to Procedure 001-027 in Section 1.

(/qs3/pubsys2/xml/en/procedures/84/84-001-027.html)

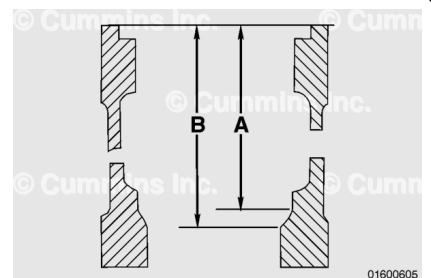


Measure the depth (position) of the lower liner bore entry chamfer from the top deck surface.

After machining the cylinder block head deck surface, all cylinder lower liner bore entry chamfers will require machining to bring the cylinder block back to original specifications.

Use the following procedure for cylinder bore concentricity specifications and machining processes. Refer to Procedure 001-027 in Section 1.

(/qs3/pubsys2/xml/en/procedures/84/84-001-027.html)

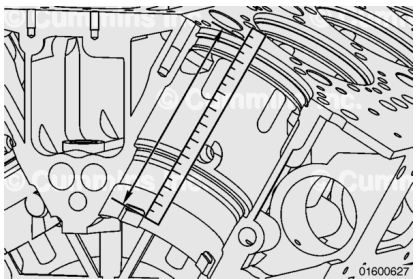


Depending on the amount of material removed from the head deck surface, the edge of the liner catchers may require machining.

Measure the distance between the edge of the liner catchers and the cylinder block head deck surface.

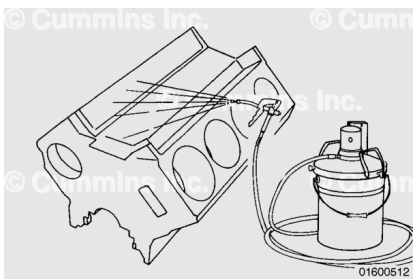
Distance Between Liner Catchers and Cylinder Block
Head Deck Surface

mm		in
288.900	MIN	11.374
289.100	MAX	11.382



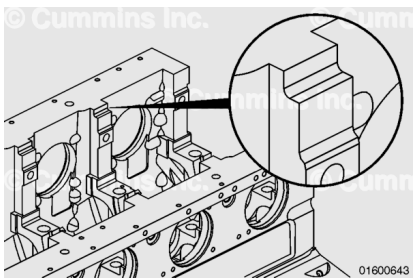
Clean the block ready for reuse as described earlier in this procedure.

Note : If the cylinder block is not to be used immediately, spray a rust preventive solution on the block and cover the cylinder block to protect it from the environment.



⚠ CAUTION ⚠

Machining of the left bank pan rail inner wall can only be done on QSK60 cylinder blocks. Do not attempt this operation on QSK45 cylinder blocks as engine damage could possibly occur.



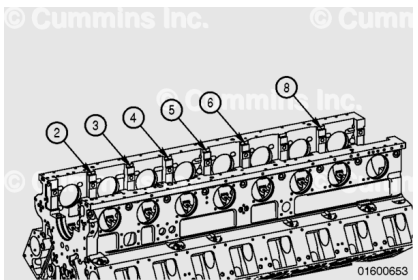
Pan Rail:

The QSK60 left bank pan rail inner wall can be machined at the main bearing cap location. The machine cuts allow for removal of the main bearing caps without the need to remove the crankshaft counterweights and installation of the main bearing caps without the need to remove the crankshaft counterweights or rotate the crankshaft.

Note : Some QSK60 engines may already have this feature machined in to the left bank pan rail inner wall.

⚠ CAUTION ⚠

Do not machine main bearing cap locations 1, 7, and 9 as engine damage could possibly occur.



⚠ CAUTION ⚠

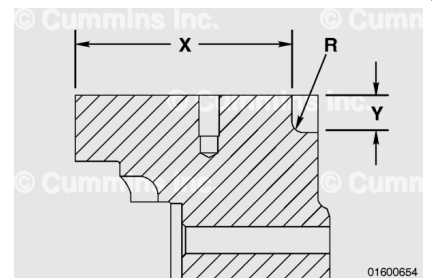
Only the left bank pan rail inner wall can be machined. Do not machine the right bank pan rail inner wall or both pan rail inner walls as engine damage could possibly occur.

The machine cuts **must only** be done at main bearing cap locations 2, 3, 4, 5, 6, and 8.

Machine the left bank pan rail inner wall in the main bearing cap locations shown in the illustration to the following dimensions and specification.

Left Bank Pan Rail Inner Wall Dimensions

	mm		in
X	109.25	MIN	4.301
	109.75	MAX	4.321
Y	16.75	MIN	0.659
	17.25	MAX	0.679
R	4.55	MIN	0.179
	5.05	MAX	0.199

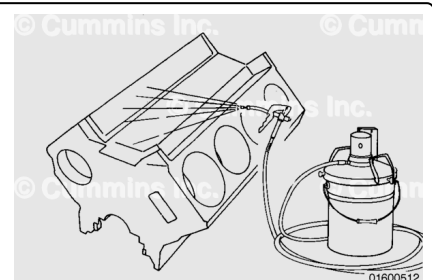


Measurements

	micro-m	in
Surface Finish	3.2	0.0001

Clean the block ready for reuse as described earlier in this procedure.

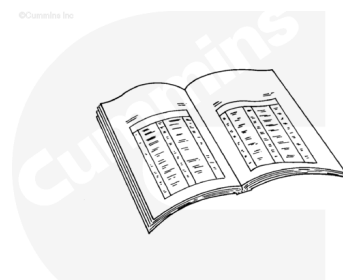
Note : If the cylinder block is **not** to be used immediately, spray a rust preventive solution on the block and cover the cylinder block to protect it from the environment.



Finishing Steps

- Install the cylinder liners. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/84/84-001-028-tr.html)

- Install the crankshaft. Refer to Procedure 001-016 in Section 1. (</qs3/pubsys2/xml/en/procedures/84/84-001-016.html>)
- Install the gear housing. Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 001-033 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-033-tr.html>) Use the following procedure in the Industrial QSK45G and QSK60G Series Engines Troubleshooting and Repair Manual, Bulletin 4021383, and the QSK60G Generator CM700 Troubleshooting and Repair Manual, Bulletin 4021396. Refer to Procedure 001-033 in Section 1. (</qs3/pubsys2/xml/en/procedures/84/84-001-033-tr.html>)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/84/84-002-004-tr.html>)
- Install the engine. Refer to Procedure 000-002 in Section 0. (</qs3/pubsys2/xml/en/procedures/84/84-000-002.html>)



ck800wa